Exponentials & Logarithms

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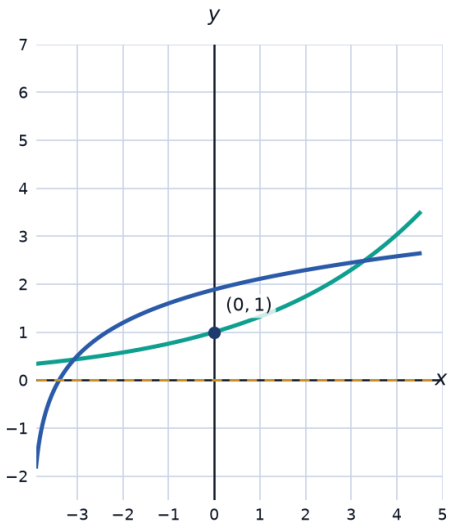
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate 2^5.

2 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate 3^{-2}.



3 **[Foundational]** • **Target: Evaluating exponential expressions**

Evaluate $16^{1/2}$.

4 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate $\log_2 32$.

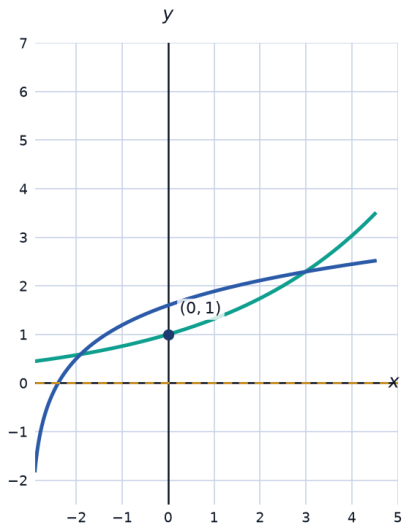
5 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate $\log_{10} 0.01$.

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6 [Foundational] • Target: Log laws: product and quotient

Write $\log_2(8x)$ as a sum of logarithms.



7 [Foundational] • Target: Log laws: product and quotient

Write $\log_3\left(\frac{x}{9}\right)$ as a difference of logarithms.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

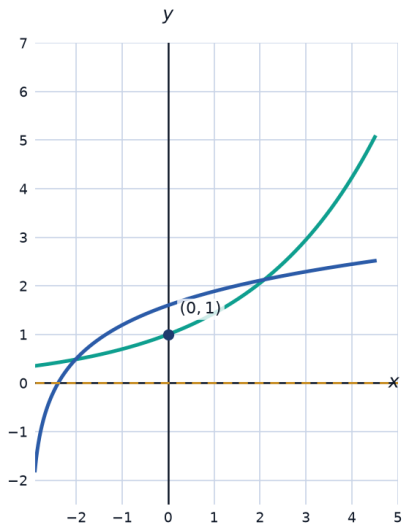
9 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.

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10 [Intermediate] • Target: Log laws: the power rule

Write $\log(x^5)$ using the power rule.



11 [Intermediate] • Target: Log laws: the power rule

Evaluate $\log_2(4^3)$ using the power rule.

12 [Intermediate] • Target: Log laws: the power rule

Write $2\log_3 x + \log_3 y$ as a single logarithm.

13 [Intermediate] • Target: Expanding and condensing logarithmic expressions

Expand $\log\left(\frac{x^3y}{z^2}\right)$ fully.

14 [Intermediate] • Target: Expanding and condensing logarithmic expressions

Condense $3\ln x - 2\ln y$ into a single logarithm.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

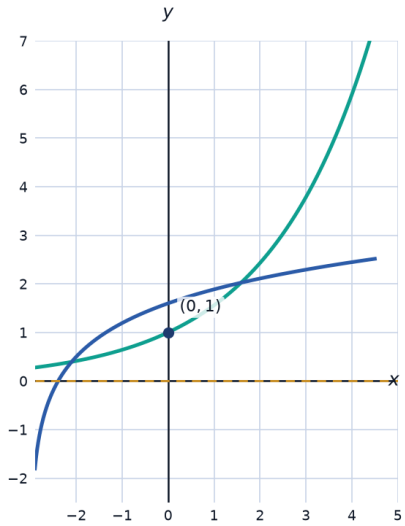
15 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 6.01 Limits & Derivatives

Expand $\log_2\left(\sqrt{x}\cdot y^3\right)$ fully.

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16 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 7.01 Integration & Applications

Solve $2^x = 32$.



17 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 9.01 Statistics & Probability

Solve $3^{x+1} = 81$.

18 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 1.01 Algebra & Equations

Solve $5^{2x} = 125$.

19 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 2.01 Functions & Transformations

Solve $2^x = 10$ for x, rounded to three decimal places.

20 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 3.01 Sequences & Series

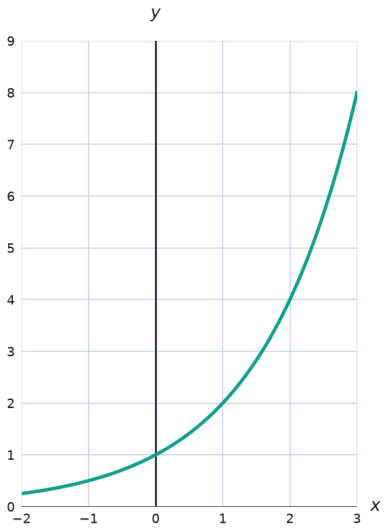
Solve $3^{2x} = 50$ for x, rounded to three decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exponential growth and decay models

The graph shows an exponential function. Based on the graph, does it represent growth or decay?



a. Neither

b. Growth

c. Cannot be determined

d. Decay

22 [Foundational] • Target: Log laws: the power rule

Write $4\log x - \log y$ as a single logarithm.

a. $\log\left(\frac{x^4}{y}\right)$

b. $\log(4x - y)$

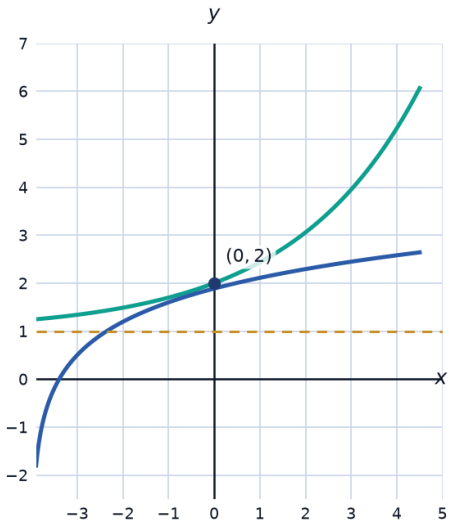
c. $4\log\left(\frac{x}{y}\right)$

d. $\log\left(\frac{4x}{y}\right)$

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23 [Foundational] • Target: Expanding and condensing logarithmic expressions

Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.



a. $\log(x \cdot y^2)$

b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c. $\log(\sqrt{x} \cdot y^2)$

d. $\log(2\sqrt{x} \cdot y)$

24 [Foundational] • Target: Solving exponential equations (matching bases)

Solve $4^x = \frac{1}{16}$.

a. $x = 4$

b. $x = -4$

c. $x = -2$

d. $x = 2$

25 [Foundational] • Target: Solving exponential equations (matching bases)

Solve $2^{2x-1} = 32$.

a. $x = 3$

b. $x = 2.5$

c. $x = 6$

d. $x = 5.5$

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26 [Foundational] • Target: Natural logarithms and e

Evaluate $\ln\left(\frac{1}{e}\right)$.

a. 1

b. e

c. -1

d. 0

27 [Foundational] • Target: Natural logarithms and e

Evaluate $e^{\ln 7}$.

a. 7

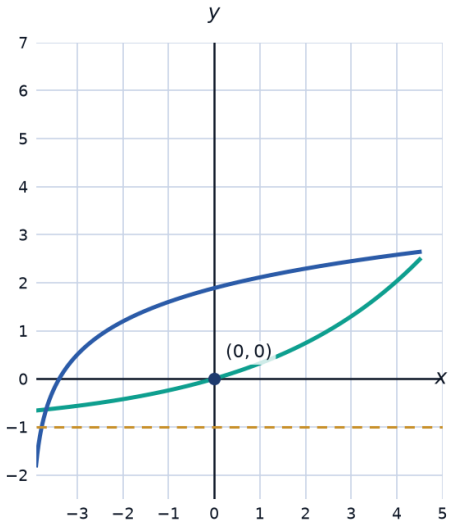
b. $\ln 7$

c. 1

d. e^7

28 [Foundational] • Target: Exponential growth and decay models

The population of a town is modeled by $P(t) = 8{,}000e^{0.025t}$, where t is in years. Find $P(15)$, rounded to the nearest whole number.



a. 8,300

b. 11,000

c. 11,640

d. 12,000

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Exponential growth and decay models

Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

a. Decay

b. Neither

c. Growth

d. Cannot be determined

30 [Intermediate] • Target: Exponential growth and decay models

Determine whether $M(t) = 30e^{0.01t}$ models growth or decay.

a. Neither

b. Cannot be determined

c. Decay

d. Growth

31 [Intermediate] • Target: Word problems: exponential models in context

An investment of 3,000 dollars grows according to $A(t) = 3{,}000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.

a. 4,200

b. 4,800

c. 4,475

d. 3,400

32 [Intermediate] • Target: Word problems: exponential models in context

A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ (°C), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

a. $38.5\,^\circ$C

b. $50.0\,^\circ$C

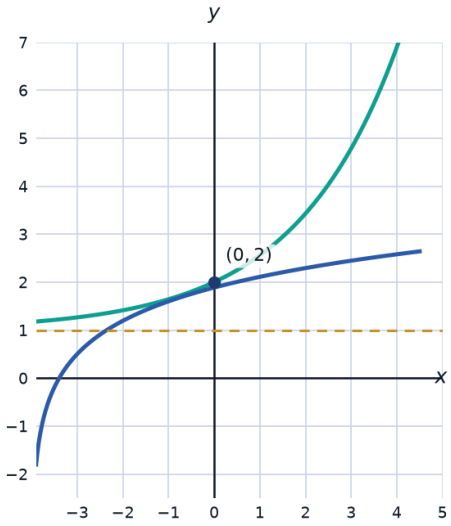
c. $43.1\,^\circ$C

d. $22.0\,^\circ$C

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33 [Intermediate] • Target: Word problems: exponential models in context

A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.



a. 4,000

b. 2,000

c. 8,000

d. 1,500

34 [Intermediate] • Target: Word problems: exponential models in context

The value of a car depreciates according to $V(t) = 25{,}000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.

a. 11,093

b. 12,750

c. 9,500

d. 10,625

35 [Intermediate] • Target: Exponentials & Logarithms

Evaluate $4^{\frac{1}{2}}$.

a. 4

b. 5

c. 2

d. 6

36 [Intermediate] • Target: Exponentials & Logarithms

In a laboratory calibration sequence, evaluate 5^2.

a. 27

b. 10

c. 23

d. 25

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37 [Intermediate] • Target: Exponentials & Logarithms

Solve $4^x = 256$.

a. 4

b. 3

c. 6

d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 1.01 Algebra & Equations

Solve $4^{x-3} = 16$.

a. 5

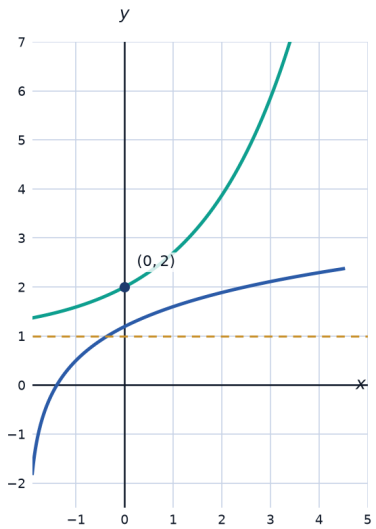
b. 6

c. 2

d. 4

39 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 2.01 Functions & Transformations

Combine $\log_2 2 + \log_2 2$ into a single logarithm and evaluate.



a. 3

b. 1

c. 4

d. 2

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40 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 3.01 Sequences & Series

Combine $\log_5 125 - \log_5 25$ into a single logarithm and evaluate.

a. 3

b. 5

c. 1

d. 2

41 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 4.01 Trigonometry

Evaluate $\log_2 (8^2)$ using the power rule.

a. 6

b. 8

c. 5

d. 4

42 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $\log_5 (x - 2) = 4$.

a. 629

b. 630

c. 627

d. 625

43 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 6.01 Limits & Derivatives

Solve $\log_5 (x) = 3$.

a. 123

b. 128

c. 125

d. 127

44 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 7.01 Integration & Applications

Evaluate $\ln(e^6)$.

a. 5

b. 7

c. 6

d. e^6

45 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 9.01 Statistics & Probability

A quantity is modeled by $N(t) = 500e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 273.0

b. 500

c. 303.3

d. 333.6

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